

Qwen3 Coder MoE — comprehensive benchmark report

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Three benches, two models, four sides on single-call plus a fifth fan-out side at N=200 subagents. All exec results come from the real vendor/bin/phpunit suite on wesleysimplicio/sistema-sindico.

Headline — single-call (4 sides)

Model	Metric	base	cli	cli+sp	cli+ag	Δcli	Δcli+sp	Δcli+ag
Qwen3-Coder-30B-A3B-Instruct	exec	33%	91%	91%	91%	+58	+58	+58
Qwen3-Coder-30B-A3B-Instruct	regex	36%	90%	98%	90%	+54	+62	+54
Qwen3-Coder-Next	exec	50%	83%	83%	91%	+33	+33	+41
Qwen3-Coder-Next	regex	44%	100%	94%	100%	+56	+50	+56

Headline — fan-out N=200

Model	fn per-att	fn modal	rx per-att	rx modal	avg uniq	wall-clock	cost
Qwen3-Coder-30B-A3B-Instruct	994/2400 (41%)	5/12	2231/2400 (92%)	11/12	6.2	520s	\$0.0000
Qwen3-Coder-Next	2208/2400 (92%)	12/12	2297/2400 (95%)	12/12	28.2	240s	\$0.0000

Key findings

1. cli + ag is the only side that beats cli reliably on functional

On the exec bench (real PHPUnit), cli + ag (verify-loop) recovers one or two cases per model that single-shot cli misses, by feeding the PHPUnit tail back as classified retry feedback. cli + sp ties or trails cli in aggregate — it adds ~1,000 tokens/call of runtime preamble that, in this single-call context, doesn't translate into pass-rate gains.

2. Regex INFLATES on Qwen3-Coder-30B-A3B-Instruct under temp=0.7

Fan-out N=200 at temp=0.7: 30B-A3B shows regex-vs-functional disagreement on 6 of 12 cases. Regex scores 100% (every pattern matched on every output) but PHPUnit exit code 0 was hit ZERO times in 200 attempts. The model produces output that LOOKS right but the runtime behaviour is wrong — exactly the criticism that 'regex doesn't mean it works'.

3. Qwen3-Coder-Next is dramatically more robust to temp=0.7

Same fan-out config, same prompts, same 12 cases — Coder-Next modal-vote passes 12/12 cases on real PHPUnit, while 30B-A3B modal-vote passes 5/12. cli's 6-layer contract is enough; Coder-Next preserves semantic correctness across temperature-induced variations where 30B-A3B drifts.

4. Some cases are model-capability ceilings, not feedback-loop problems

password_require_symbol failed for every side and every iteration on 30B-A3B (3-attempt verify-loop exhausted; 200 fan-out attempts exhausted). It only passed on Coder-Next, and only with modal-vote fan-out at 187/200. This is a model-capability boundary — no amount of feedback or retry helps; either the model can or it can't.

Cost & latency (exec bench, per call)

Model	Side	tokens/call	ms/call	notes
Qwen3-Coder-30B-A3B-Instruct	baseline	1259	5030	
Qwen3-Coder-30B-A3B-Instruct	cli	1449	4983	
Qwen3-Coder-30B-A3B-Instruct	cli+sp	2412	4879	+sp runtime ~1k tokens
Qwen3-Coder-30B-A3B-Instruct	cli+ag	1579	5332	1.17 avg attempts
Qwen3-Coder-Next	baseline	1280	3571	
Qwen3-Coder-Next	cli	1449	3485	
Qwen3-Coder-Next	cli+sp	2416	3398	+sp runtime ~1k tokens
Qwen3-Coder-Next	cli+ag	1858	4210	1.17 avg attempts

Fan-out per-task (N=200, modal-vote)

Format per cell: **fn% / rx% / fn-modal / unique-outputs**. P = modal pass, . = fail. uX = X unique outputs out of 200.

Task	Q3C-30B-A3B-Instruct	Q3C-Next
password_strength	97/100/P/u5	100/100/P/u5
password_require_symbol	0/ 15/. /u9	93/ 99/P/u13
env_get_int	0/100/. /u7	99/100/P/u41
env_get_bool	0/100/. /u6	49/ 49/P/u28
admin_only_allowed_roles	0/100/. /u1	100/100/P/u2
rate_limit_bucket_key	0/100/. /u1	100/100/P/u5
base_repository_build_where_sql	0/100/. /u2	100/100/P/u15
router_has	0/100/. /u2	100/100/P/u12
bugfix_password_policy_lowercase	100/100/P/u9	100/100/P/u22
password_assess	100/100/P/u6	100/100/P/u6
base_repository_build_update_sql	100/100/P/u17	61/100/P/u139
router_extract_params	100/100/P/u10	100/100/P/u50

Exec per-task × model × side

Format per cell: **base / cli / cli+sp / cli+ag(attempts)**. P = real PHPUnit suite green; . = fail.

Task	Q3C-30B-A3B-Instruct	Q3C-Next
password_strength	./P/P/P(1)	./P/P/P(1)
password_require_symbol	./././.(3)	./P/./P(1)
env_get_int	./P/P/P(1)	P/P/P/P(1)
env_get_bool	./P/P/P(1)	././P/P(1)
admin_only_allowed_roles	P/P/P/P(1)	P/P/P/P(1)
rate_limit_bucket_key	./P/P/P(1)	./P/P/P(1)
base_repository_build_where_sql	./P/P/P(1)	./P/P/P(1)
router_has	./P/P/P(1)	P/P/P/P(1)
bugfix_password_policy_lowercase	P/P/P/P(1)	P/P/./P(1)
password_assess	P/P/P/P(1)	P/P/P/P(1)
base_repository_build_update_sql	./P/P/P(1)	././P/.(3)

router_extract_params	P/P/P/P(1)	P/P/P/P(1)
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